Thong (Tom) Huynh

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Education & Awards

Ontario Tech University	B.Eng., Mechatronics Engineering (GPA: 4.25/4.3)	2024–2028
Competitions	International Youth Robot Competition 2019 (1st place), MakeX (Innovative Design), FIRST Robotics	
Awards	Undergraduate Research Award, STAR Award, Merit Award, President's List	

Skills

CAD and Design:	**SolidWorks Certified Professional**, Siemens NX, 3D-Printing, GD&T, DFM/DFA.
Robotics Hardware:	NVIDIA Jetson Orin Nano/AGX, Raspberry Pi, LiDAR, IMUs, sensors, actuators.
Robotics and Simulation:	ROS 2, Gazebo, NVIDIA Isaac Sim/Lab, Nav2.
Programming and Vision:	Linux, Python, C++, Git, OpenCV, PyTorch, YOLO, RF-DETR, DINO/DINOv2.

Experience

Robotics Software Engineer | ABI Ltd. *May 2026 – Present*

- Trained a PPO policy in **NVIDIA Isaac Lab** with dough-geometry domain randomization for sim-to-real robotic bread scoring.
- Developed high-fidelity digital twins of industrial robotic workcells in **NVIDIA Isaac Sim** to accelerate system validation, reducing on-site commissioning time by 30%.
- Architected an in-house, two-stage YOLO with CNN head for a bakery inspection pipeline (1M+ pieces/day), achieving < 100 ms inference latency and cutting $60K in outsourcing costs.
- Engineered an internal computer vision tooling suite deployed at 2 customer sites and adopted company-wide across the engineering team, improving model evaluation and deployment workflows.

Research Assistant | MARS Lab *May 2025 – May 2026*

- Designed a custom **4-bar linkage folding mechanism** and 10+ 3D-printed mechanical components (SLS nylon and carbon fiber) to maximize stiffness-to-weight ratio and increase payload capacity by 20%.
- Formulated forward/inverse kinematics for quadruped locomotion and built an **end-to-end Gazebo simulation pipeline** to validate motor torque and gait behavior.
- Implemented OctoMap-based 3D mapping, navigation, and obstacle avoidance with PX4 & Jetson Nano via MAVROS.

Design Team Lead | Ontario Tech RoboMaster *September 2024 – Present*

- Directed the mechanical design of a 500+ component competition robot in SolidWorks with **full DFM/DFA compliance** for CNC machining, sheet metal, and 3D printing.
- Engineered a 2-DOF turret with belts and herringbone gears, reducing backlash and improving the accuracy of the shooter by 27%.
- Performed **failure analysis and FEA**, managed CAD version control, standardized team processes for seamless design across subsystems.
- Led a team of 6+ team members in mechanical design, **GD&T**, engineering drawings, and robotics design principles.

Projects

SensQ – Autonomous Mobile Robot Platform *January 2026 – Present*

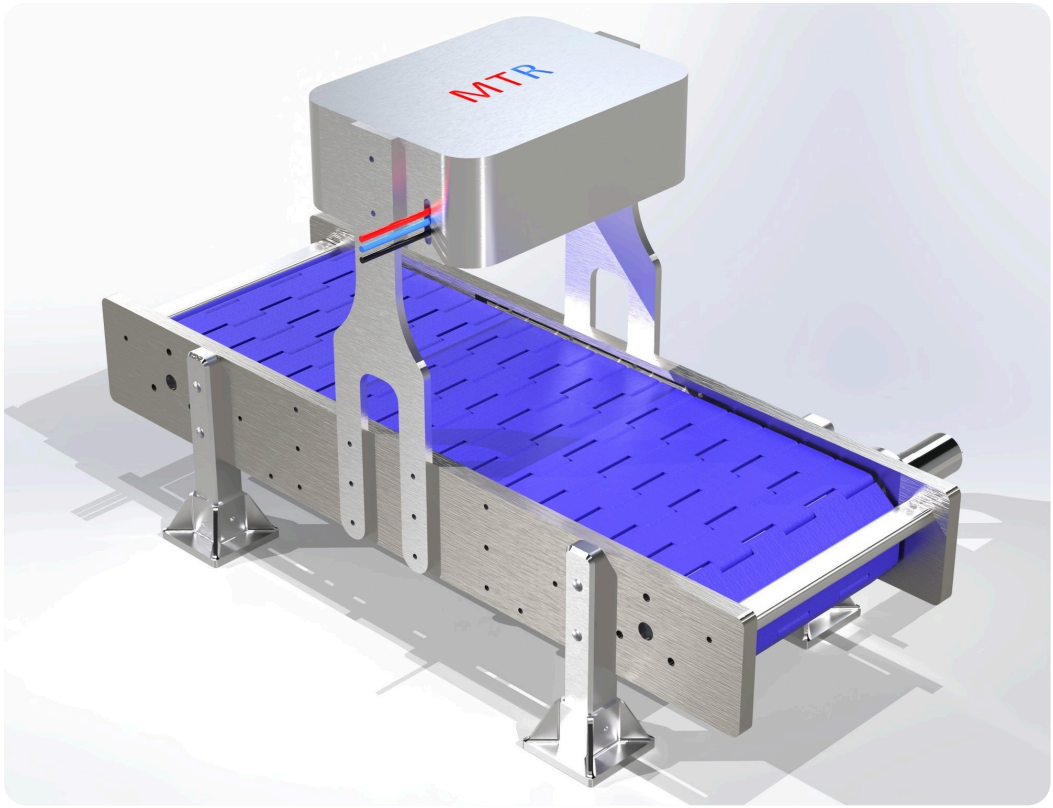
- Designed and 3D-printed an AMR with a custom suspension mechanism, achieving speeds of 0.7 m/s with sub-2mm repeatability in point-to-point navigation tasks.
- Simulated system performance in Gazebo and integrated end-to-end physical hardware with ROS 2.
- Implemented Nav2 and SLAM Toolbox navigation, custom A* and Boustrophedon coverage planning, and AprilTag-based docking.

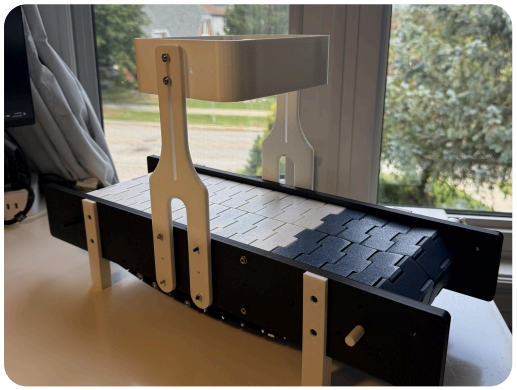
Bolt Eye – Automated Inspection System *August 2025 – Present*

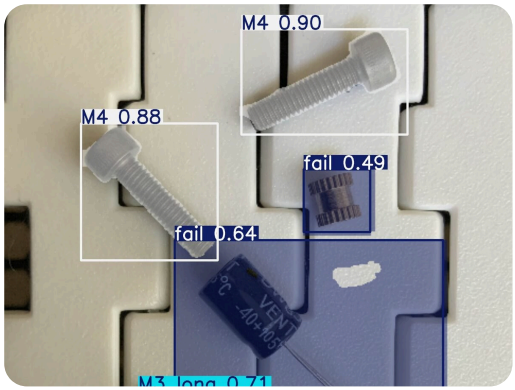
- Built a 1 m/s motorized conveyor with encoder tracking, belt tensioner, and synchronized vision acquisition for real-time inspection.
- Developed a real-time defect detection and classification pipeline using YOLOv11 and DINOv2, achieving sub-100 ms inference latency.
- Detected and classified surface defects as small as 2 mm on an active production-line conveyor operating at 1 m/s.

6-DOF Robotic Arm *May 2024 – January 2026*

- Designed and calculated joint torque profiles, gear reductions, and preload compensation across 6 degrees of freedom, increasing structural payload capacity to 2 kg while maintaining structural rigidity.
- Customized kinematics calculation with L-BFGS-B path planning for pick and place tasks with 2mm accuracy.
- Implemented with Python, MoveIt, and RViz tools for motion visualization and validation.







Bolt Eye

Automated Inspection System

A conveyor vision cell that counts, sorts and quality-checks screws, bolts and nuts as they pass.

HIGHLIGHTS

- End-to-end design, fully 3D-printed.
- Real-time detection and segmentation.
- Sorts parts by size and shape with millimetre accuracy.

SKILLS

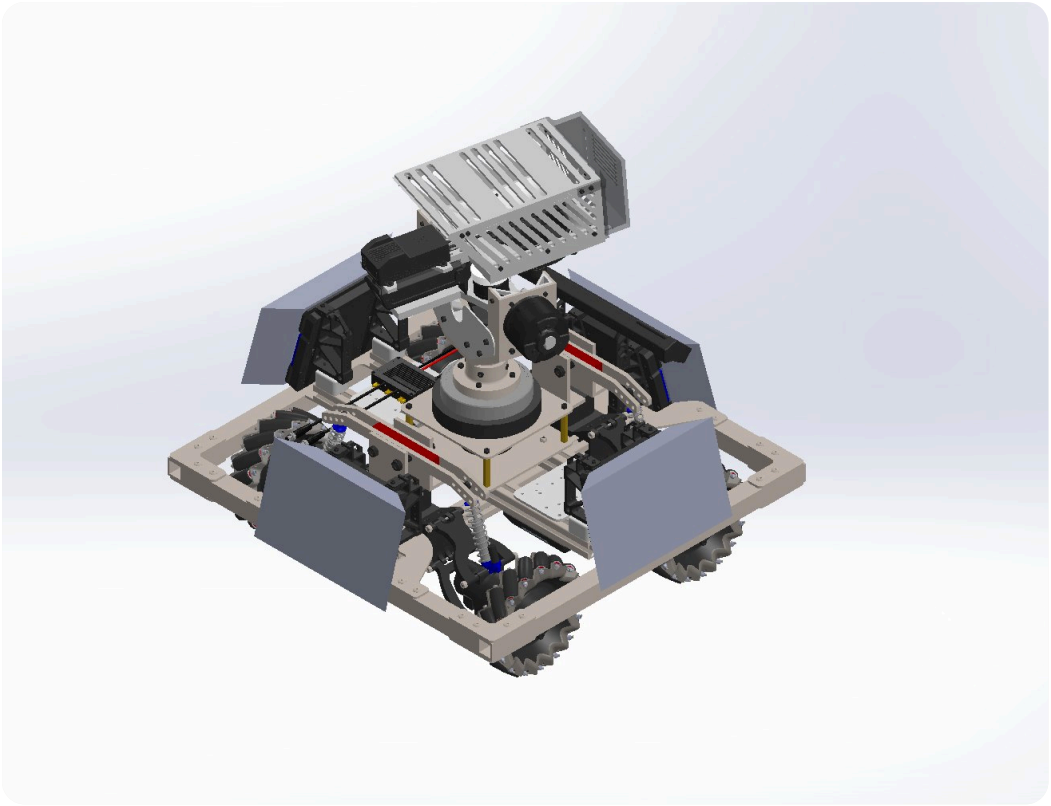
Computer vision · 3D printing · Real-time systems · CAD design

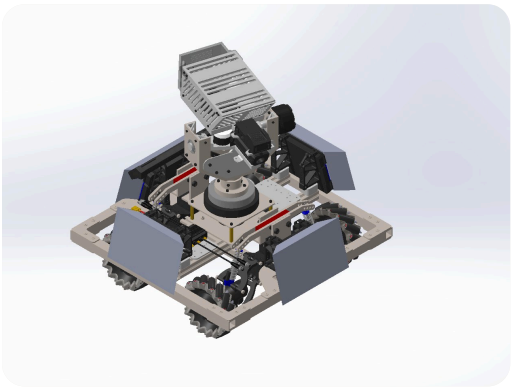
TOOLS

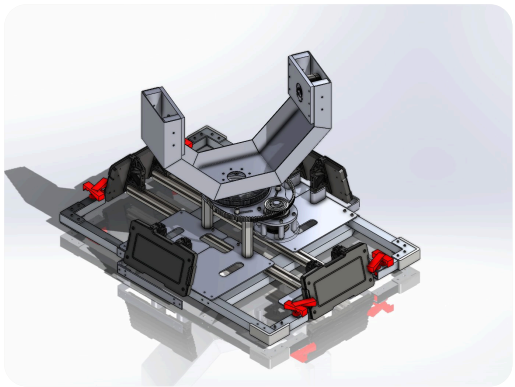
YOLOv11 · DINOv2 · OpenCV · PyTorch · SolidWorks · Jetson Nano

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RoboMaster Robot

Omnidirectional Chassis and Turret

A robot built for combat in the RoboMaster North America (ARC) competition.

HIGHLIGHTS

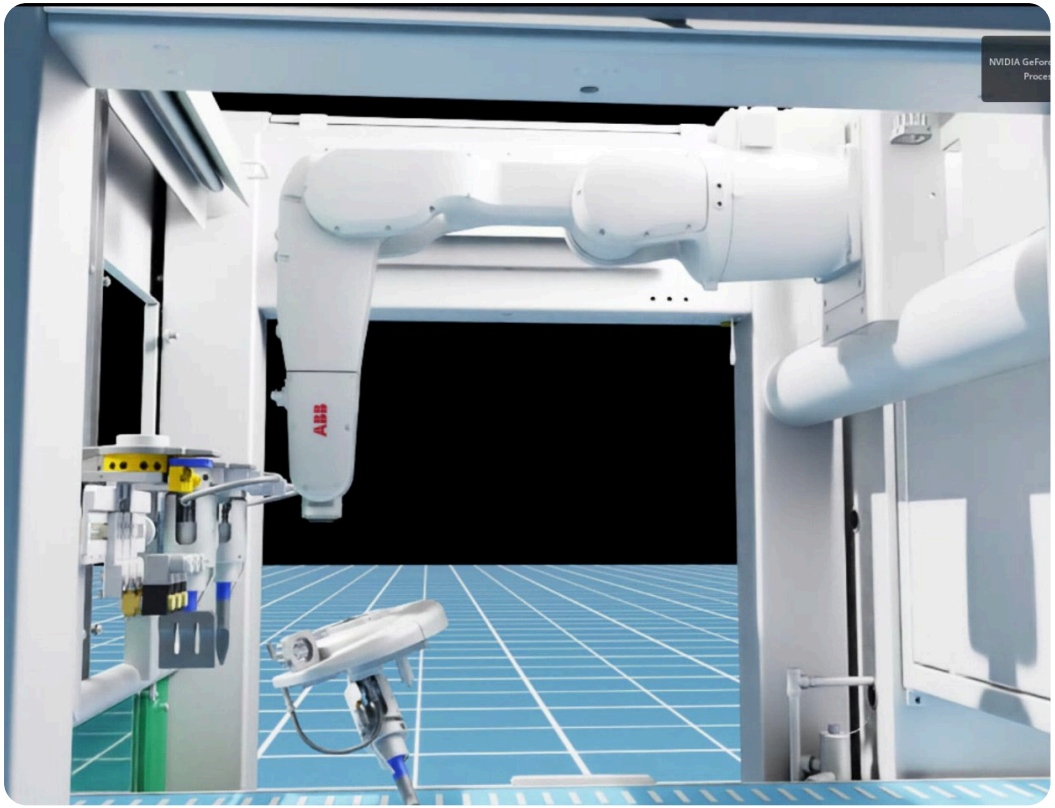
- Mechanical design directed end to end in SolidWorks, DFM/DFA compliant.
- Custom turret: timing-belt yaw and herringbone-gear pitch, minimizing backlash for accuracy.

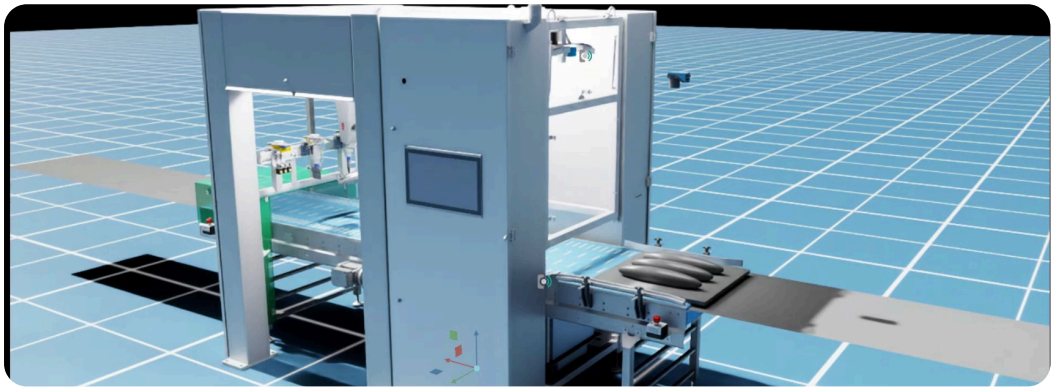
SKILLS

Mechanical design · GD&T · Design for manufacture · Design leadership

TOOLS

SolidWorks · GD&T · DFM/DFA · Engineering drawings





Digital Twin Work Cell

Isaac Sim Validation Pipeline

A virtual copy of a robotic bread-scoring cell, so layouts and motions are proven before any hardware moves.

HIGHLIGHTS

- Arm, conveyor and sensors simulated closely to the real cell.
- Robot motion and system behavior tested entirely in simulation.
- Significantly reduced commissioning time.

SKILLS

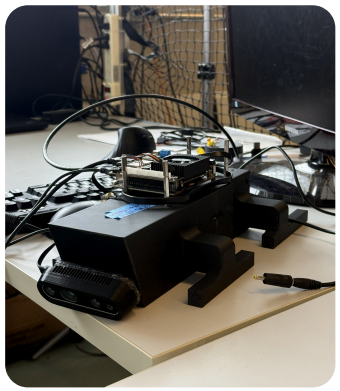
Robotics simulation | Industrial communication protocols

TOOLS

NVIDIA Isaac Sim | RobotStudio | ABB IRB1200 | ROS 2 | Python









Pegasus

Hybrid Multimodal Quadruped

A research robot that combines walking and flying on one platform.

HIGHLIGHTS

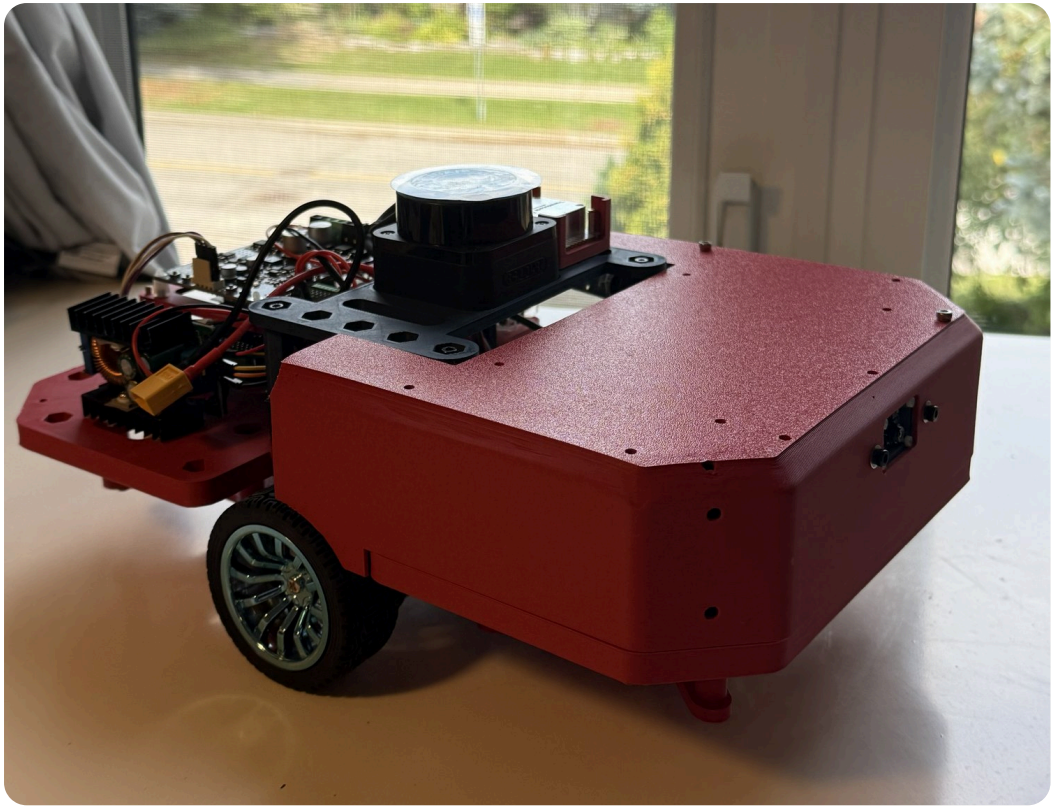
- Increased payload over the original robot by 15% with custom trot kinematics.
- Designed custom sensor mounts.
- Simulated in Gazebo for behavior validation.

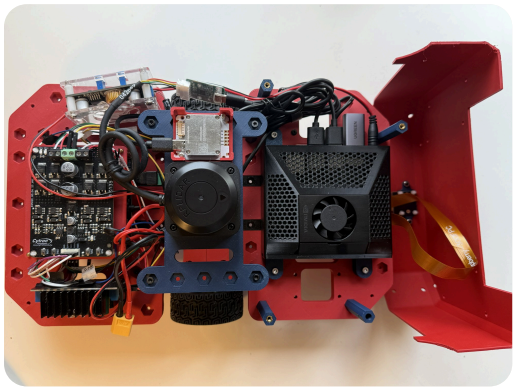
SKILLS

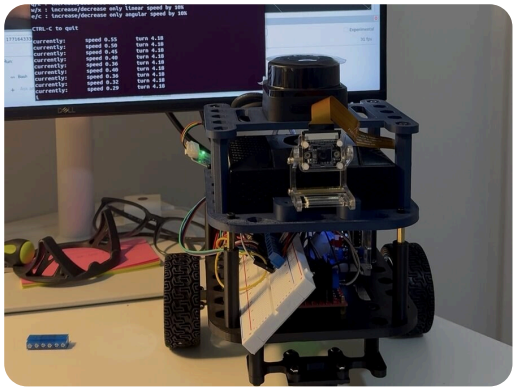
Legged locomotion · Mechanical design · Kinematics · 3D mapping

TOOLS

ROS · Gazebo · PX4/MAVROS · OctoMap · SolidWorks







SensQ

Autonomous Mobile Robot Platform

A compact robot that maps a space, covers every part of it, and finds its own way back to dock.

HIGHLIGHTS

- Chassis and hardware 3D-printed and assembled in-house.
- Coverage path planning with A* and Boustrophedon cell decomposition.
- Docking and alignment with AprilTags.

SKILLS

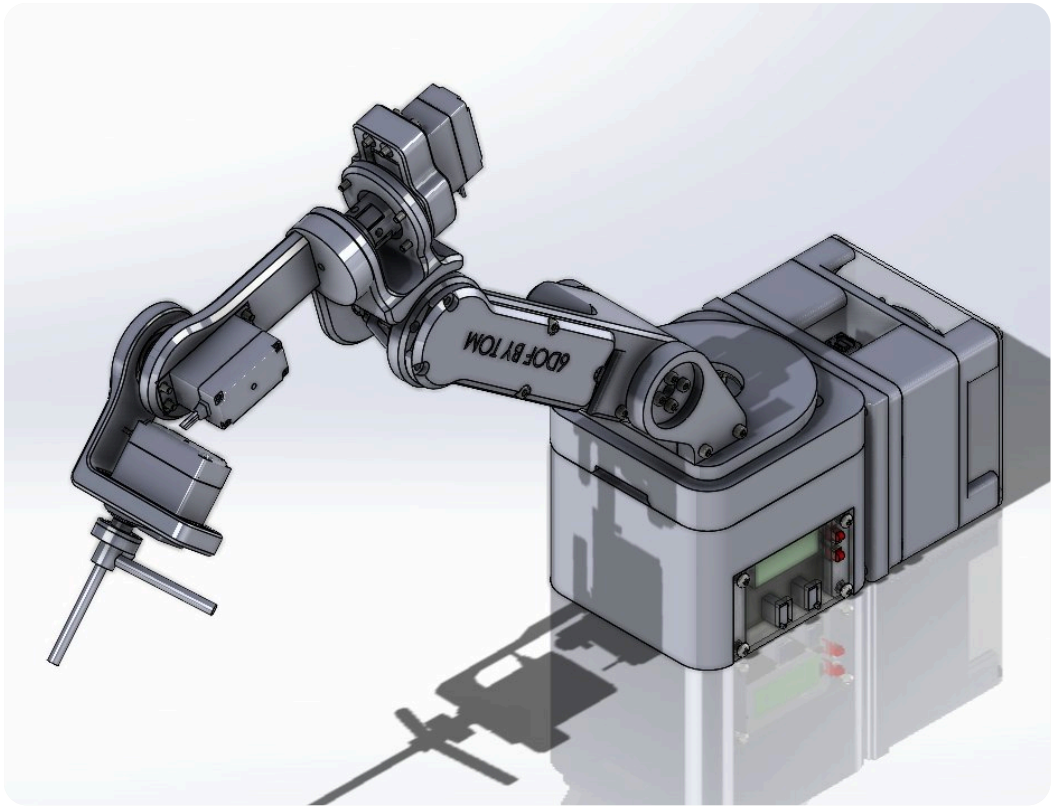
Mobile robotics · SLAM · Path planning · Hardware integration

TOOLS

ROS 2 · Nav2 · SLAM Toolbox · AprilTag · Python

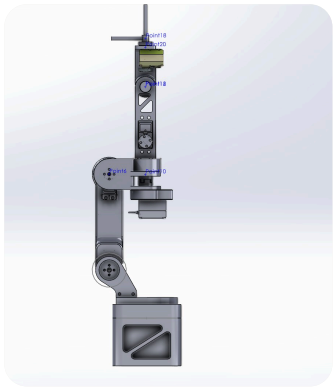
Thong Huynh

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6-DOF Robotic Arm

Pick and Place Robot

A six-axis robot arm for pick-and-place tasks with high precision and repeatability.

HIGHLIGHTS

- Designed in SolidWorks and fully 3D-printed, with a 2 kg payload and minimal backlash.
- Path planning with L-BFGS-B optimization for smooth, reachable motion.
- Simulated in Gazebo to preview motion.

SKILLS

Robot kinematics · Optimization · Motion planning · CAD design

TOOLS

ROS · RViz · Python · L-BFGS-B · SolidWorks